

Alexandra G. Guerrero

(née Alexandra G. Hanselman)

Department of Physics, University of Chicago

aghanselman@uchicago.edu • ORCID: [0000-0002-8304-0109](https://orcid.org/0000-0002-8304-0109)

EDUCATION

UNIVERSITY OF CHICAGO

SEPTEMBER 2020 – PRESENT

PhD Candidate in Physics (expected June 2027)

MS in Physics received June 2023

- Cumulative GPA: 3.9/4.0
- Advisor: Dr. Daniel Holz

MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT)

AUGUST 2016 – JUNE 2020

S.B. May 2020

- Cumulative GPA: 4.7/5.0
- Physics major, astronomy minor
- Research Advisor: Dr. Scott Hughes
- Thesis: “Investigating the influence of spin-curvature coupling on extreme mass-ratio inspirals”

FELLOWSHIPS

ILLINOIS ARCS SCHOLAR

2022 – 2025

With support from the Brinson Foundation

MCCORMICK GRADUATE FELLOWSHIP

2020 – 2021

University of Chicago

AWARDS & HONORS

Midwest Relativity Meeting Blue Apple Award

November 2024

For best student talk

Alan H. Barrett Prize

June 2020

For outstanding research by a student in Astrophysics

MIT

NASA Goddard Space Flight Center Swoosh Award for Science

July 2018

Top 3 science intern poster

PUBLICATIONS

D.B. Maclean, P. Agrawal, K. Breivik, **A.G. Guerrero**, M. Zevin, M. Renzo, C.L. Rodriguez, “Twin Peaks: Resolving Features in the Binary Black Hole Mass Function with COSMIC-METISSE,” arXiv:2606.28506 (2026)

A.G. Guerrero & D.E. Holz, “A simple model for extracting astrophysics from black hole images,” arXiv:2602.12195 (2026)

M. VanWyngarden, M. Fishbach, A. Vijaykumar, **A.G. Guerrero**, D.E. Holz, “How low can you go: Constraining the effects of catalog incompleteness on dark siren cosmology,” arXiv:2511.04786 (2025)

A.G. Hanselman, A. Vijaykumar, M. Fishbach, D.E. Holz, “Gravitational-wave dark siren cosmology systematics from galaxy weighting,” *ApJ*, **979**, 9 (2025)

A. Vijaykumar, **A.G. Hanselman**, M. Zevin, “Consistent eccentricities for gravitational wave astronomy: Resolving discrepancies between astrophysical simulations and waveform models,” *ApJ*, **969**, 132 (2024)

L.V. Drummond, P. Lynch, **A.G. Hanselman**, D.R. Becker, S.A. Hughes, “Extreme mass-ratio inspiral and waveforms for a spinning body into a Kerr black hole via osculating geodesics and near-identity transformations,” *Phys. Rev. D* **109**, 064030 (2024)

INVITED TALKS

Can we infer the metallicity distribution of (isolated) BBH progenitors?	April 2026
<i>The bleeding edge of gravitational-wave source populations with GWTC-6</i>	<i>CITA</i>

CONTRIBUTED POSTERS & PRESENTATIONS

Comparison of binary compact object merger populations from population-synthesis codes	March 2026
<i>APS Global Physics Summit</i>	

Comparison of binary compact object merger populations from population-synthesis codes	October 2025
<i>Midwest Relativity Meeting</i>	

Gravitational-wave dark siren cosmology systematics from galaxy weighting	July 2025
<i>Prospects in Theoretical Physics (PiTP) 2025: Gravitational Waves from Theory to Observation</i>	<i>(Poster)</i>

A simple model for extracting astrophysics from black hole images	November 2024
<i>Midwest Relativity Meeting</i>	<i>(Blue Apple Award winner)</i>

Gravitational-wave dark siren cosmology systematics from galaxy weighting	October 2024
<i>ARCS Foundation Illinois Chapter Scholar Awards Poster Session</i>	<i>(Poster)</i>

Gravitational-wave dark siren cosmology systematics from galaxy weighting	April 2024
<i>April APS Meeting</i>	

Estimating systematic bias in gravitational-wave dark siren cosmology measurements	October 2023
<i>ARCS Foundation Illinois Chapter Scholar Awards Poster Session</i>	<i>(Poster)</i>

Estimating systematic bias in gravitational-wave dark siren cosmology measurements	April 2023
<i>April APS Meeting</i>	

Extracting astrophysics from black hole images	November 2022
<i>ARCS Foundation Illinois Chapter Scholar Awards Poster Session</i>	<i>(Poster)</i>

Extracting astrophysics from black hole images	April 2022
<i>April APS Meeting</i>	

Constructing extreme mass ratio inspiral waveforms from harmonics: Part 2 August 2019
Summer MKI undergraduate research forum

Characterizing QPD to measure test mass position and orientation in LISA August 2018
NASA summer intern poster session (Poster; NASA Swoosh Award winner)

TEACHING EXPERIENCE

Teaching Assistant

University of Chicago

Undergraduate Experimental Physics II-III	Spring 2024, Winter-Spring 2026
Undergraduate Intermediate Classical Mechanics	Autumn 2025
Undergraduate Spacetime and Black Holes	Autumn 2021
Undergraduate Introductory Wave, Optics, and Heat	Spring 2021, 2025
Undergraduate Introductory Electricity and Magnetism	Winter 2021, 2022, 2024, 2025
Undergraduate Introductory Classical Mechanics	Autumn 2020

- Tasks include running labs and discussion sections, holding office hours, and grading

Teaching Assistant

Massachusetts Institute of Technology

Undergraduate introductory classical mechanics	January 2020
--	--------------

- Assisted in answering questions during lectures

OUTREACH

Astronomy Conversations

Volunteer

June 2024 – Present
Adler Planetarium

Recurring program where researchers engage with museum visitors and have open dialogues about science questions

Space Explorers

Summer Institute Volunteer

August 2023
University of Chicago

One week camp that advances science education for underrepresented minority high school students located around Chicago. Assisted students build a microwave telescope.

Warrior-Scholar Project

STEM Research Project Leader

August 2022
University of Chicago

One week college-preparatory boot camp for veterans and active-duty service members to aid in the shift to academia.
Project title: “Finding Earth 2.0: Exploring Exoplanets”

Lifelong Learning

Volunteer Speaker

December 2021
Virtual

Public lecture series for a senior audience from communities around Chicago.
Presentation title: “Staring into the void – how we ‘see’ a black hole”

Society of Women in Physics (SWiP) – Women and Gender Minorities in Physics (WAGMIP) Peer Mentorship Program
Organizer and Mentor

October 2021 – Present
University of Chicago

Mentorship program between graduate and undergraduate students, typically women and gender minorities, to provide support and advice throughout the year.

SERVICE

Peer Reviewer:

2026 – Present

- *The Astrophysical Journal*

Local Organization:

- Midwest Relativity Meeting University of Chicago; November 2023

Undergraduate Associate Advisor

September 2019 – June 2020

Matched with faculty advisor and provided a student perspective on academics and student life to assist and support first-year students.

MIT

Physics Department Grader

2018-2020

Graded problem sets for mathematically-advanced physics class on introduction to classical mechanics; proctored for physics advanced standing exams.

MIT

ACTIVITIES

Prospects in Theoretical Physics (PiTP) 2025: Gravitational Waves from Theory to Observation

July 2025

- Two-week summer program covering broad topics including gravitational-wave astrophysics, data analysis, waveform modeling, numerical relativity, parameter estimation, and pulsar timing arrays

LISA Data Analysis Workshop

January 2025

- Two-day workshop covering broad topics in LISA data analysis

NBIA Summer School on Gravitational Wave Astrophysics

August 2021

- Week-long school covering a large range of subjects on gravitational wave astrophysics with morning lessons and afternoon hands-on exercises

AFFILIATIONS

- LIGO Scientific Collaboration: Member
- American Physical Society (APS): Member

March 2022 – Present

December 2021 – Present